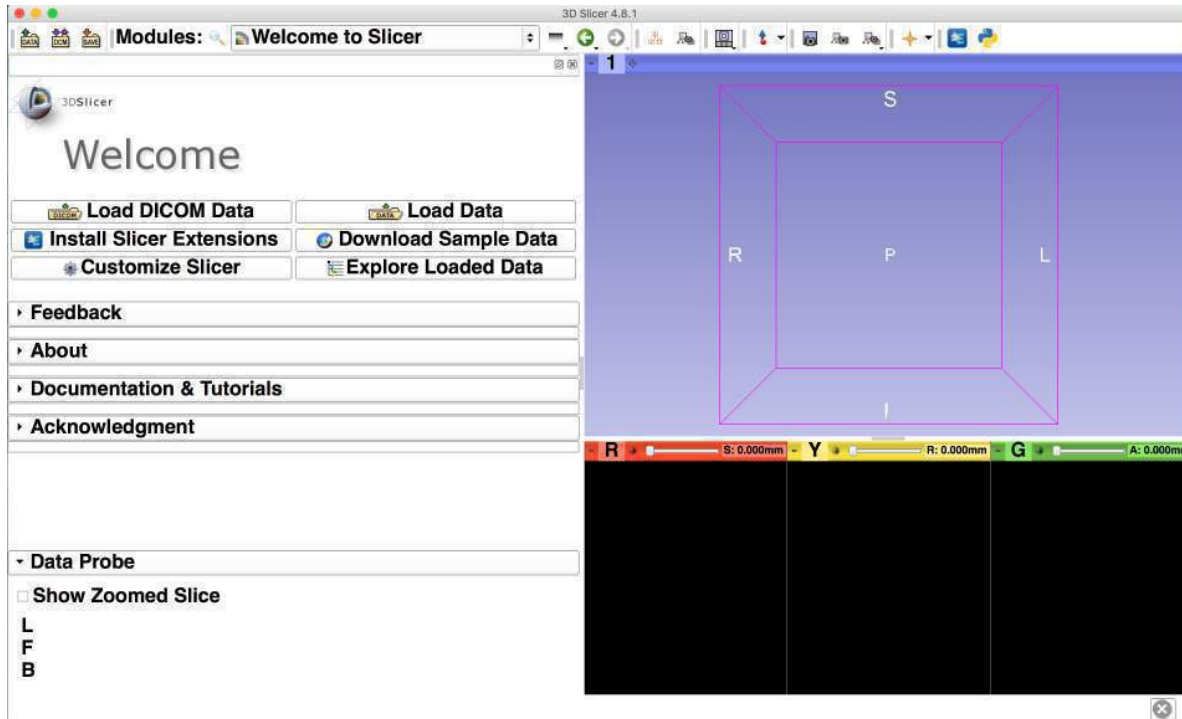


Slicer Welcome

Sonia Pujol, Ph.D.

Assistant Professor of Radiology
Brigham and Women's Hospital
Harvard Medical School

Goal

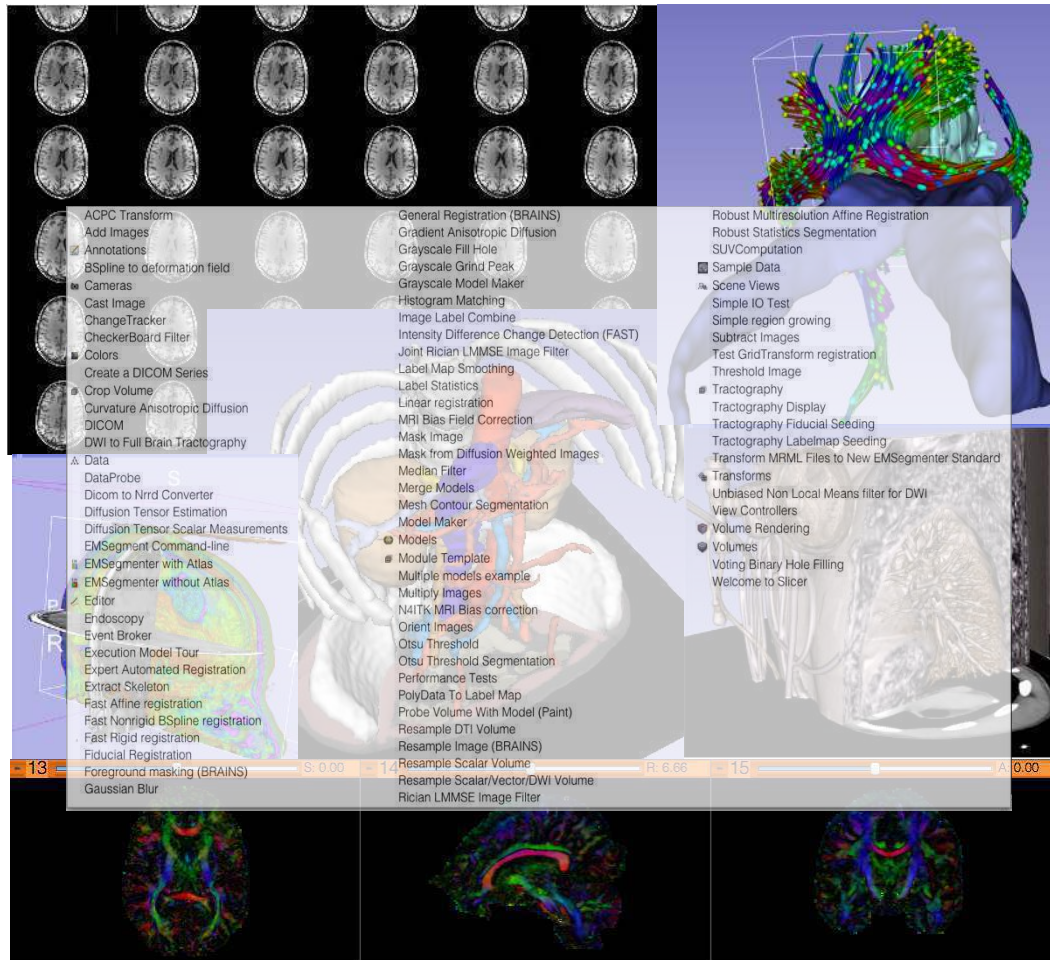


This tutorial is a short introduction to the Welcome module of the Slicer open-source software.

Slicer4 Basics

- Slicer is an open-source software for segmentation, registration and visualization of medical imaging data
- The platform is developed through a multi-institution effort of several NIH funded large-scale consortia.
- Slicer is for medical research only, and is not FDA approved

Slicer4 Basics

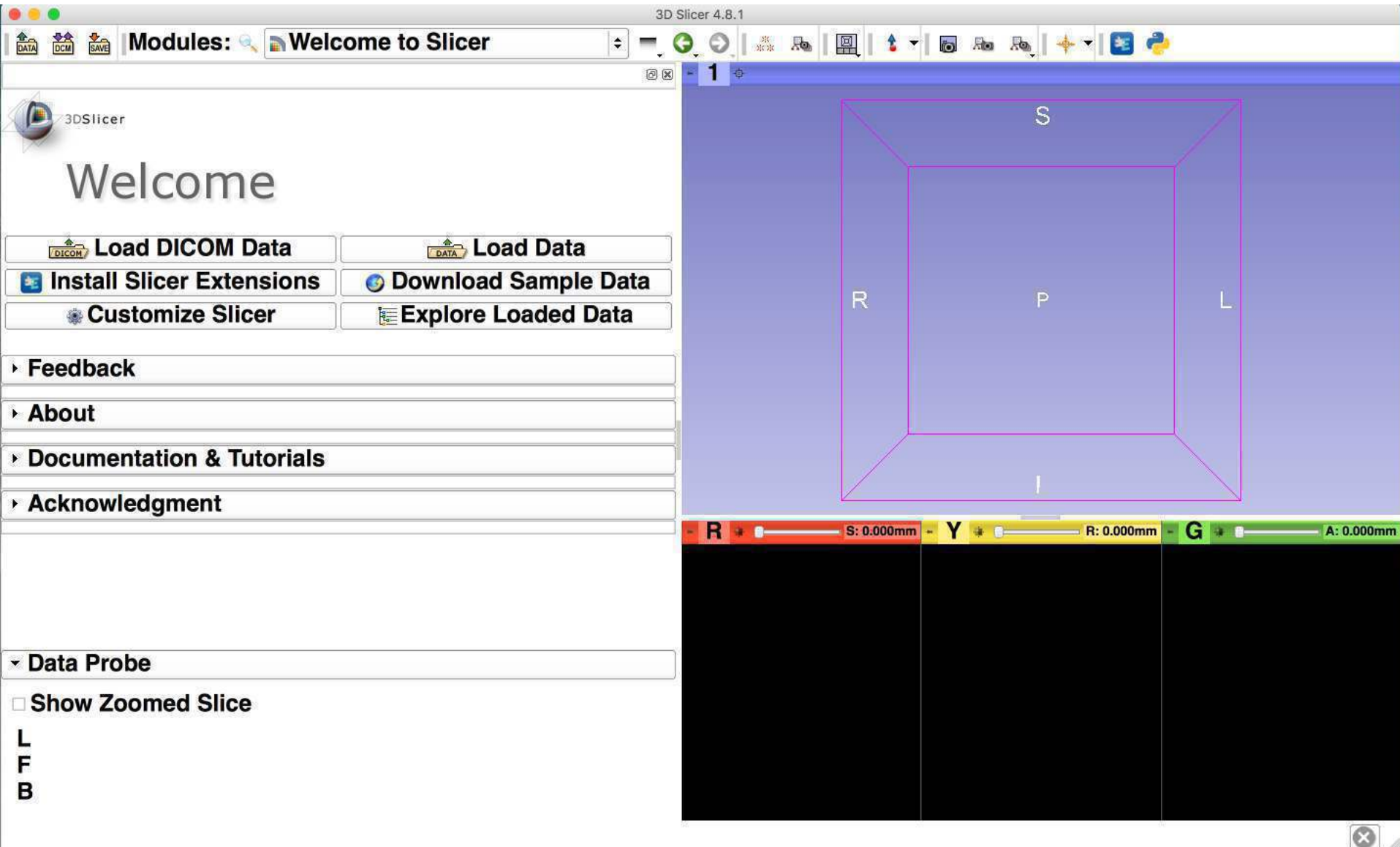


3D Slicer 4 version 4.8.1 includes 130 modules and 74 extensions for image segmentation, registration and 3D visualization of medical imaging data.

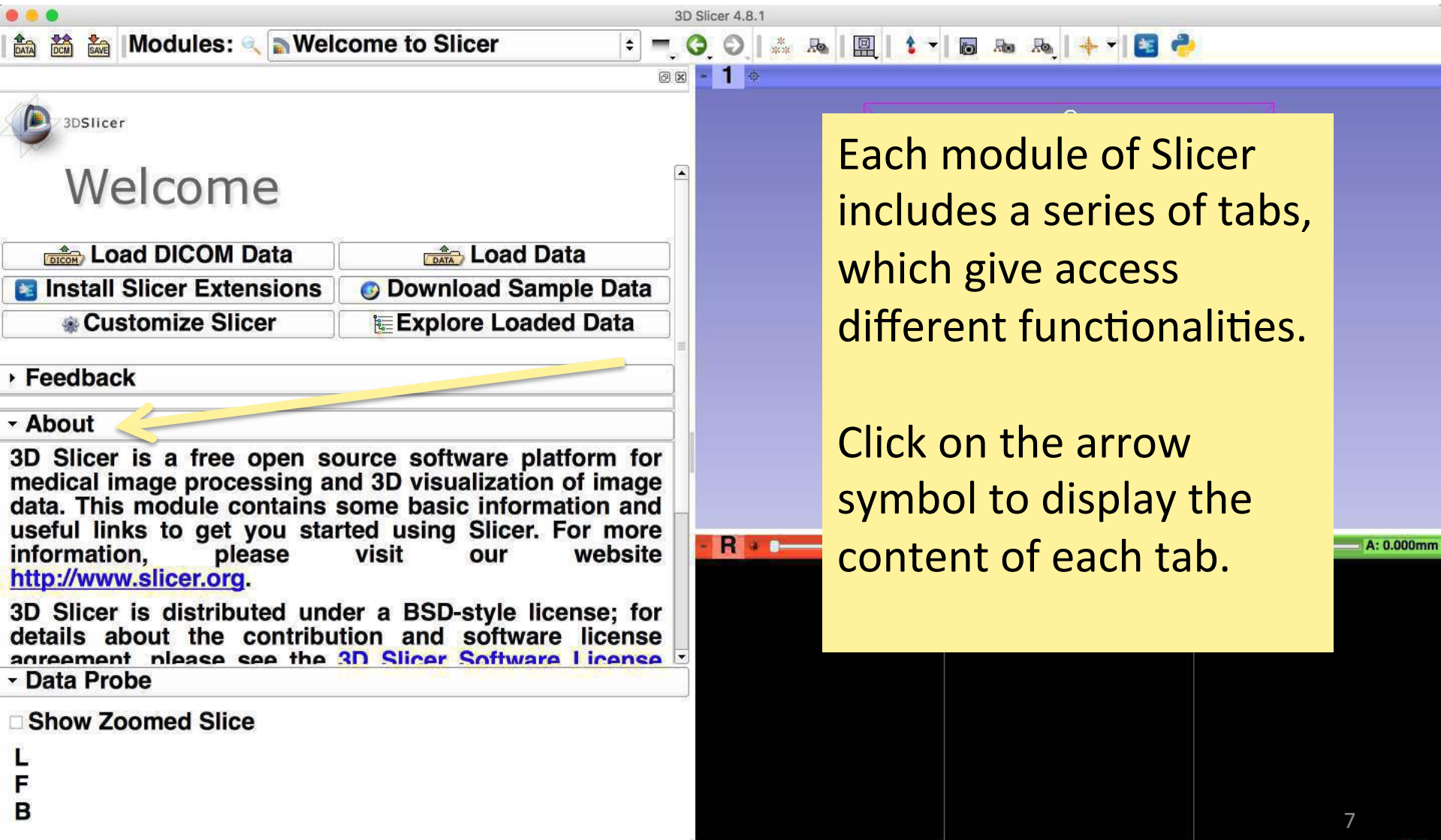
Supported Platforms

- Slicer is a multi-platform software developed and maintained on Mac OSX, Linux and Windows.
- Slicer requires a minimum of 2 GB of RAM and a dedicated graphic accelerator with 64 MB of on-board graphic memory.

3D Slicer version 4.8

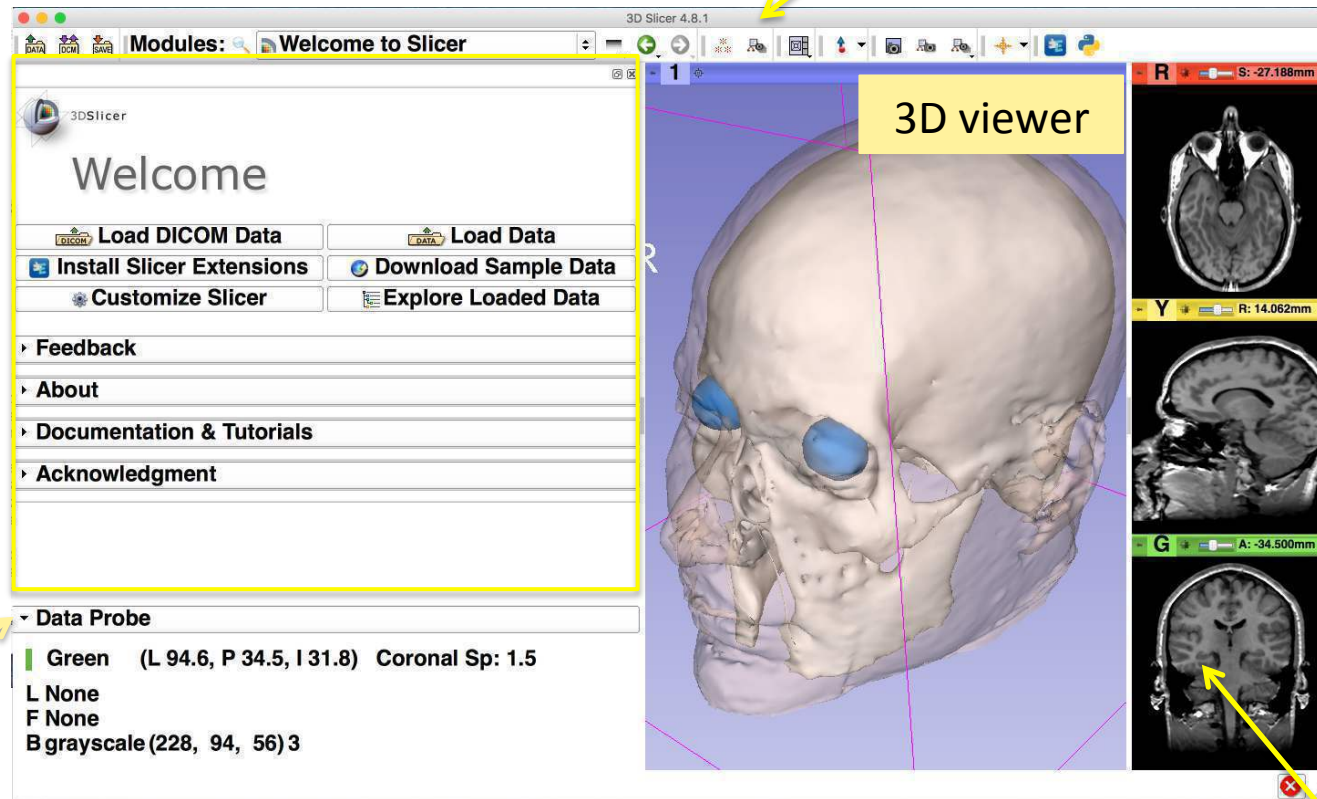


Welcome to Slicer



Slicer User Interface

Toolbar



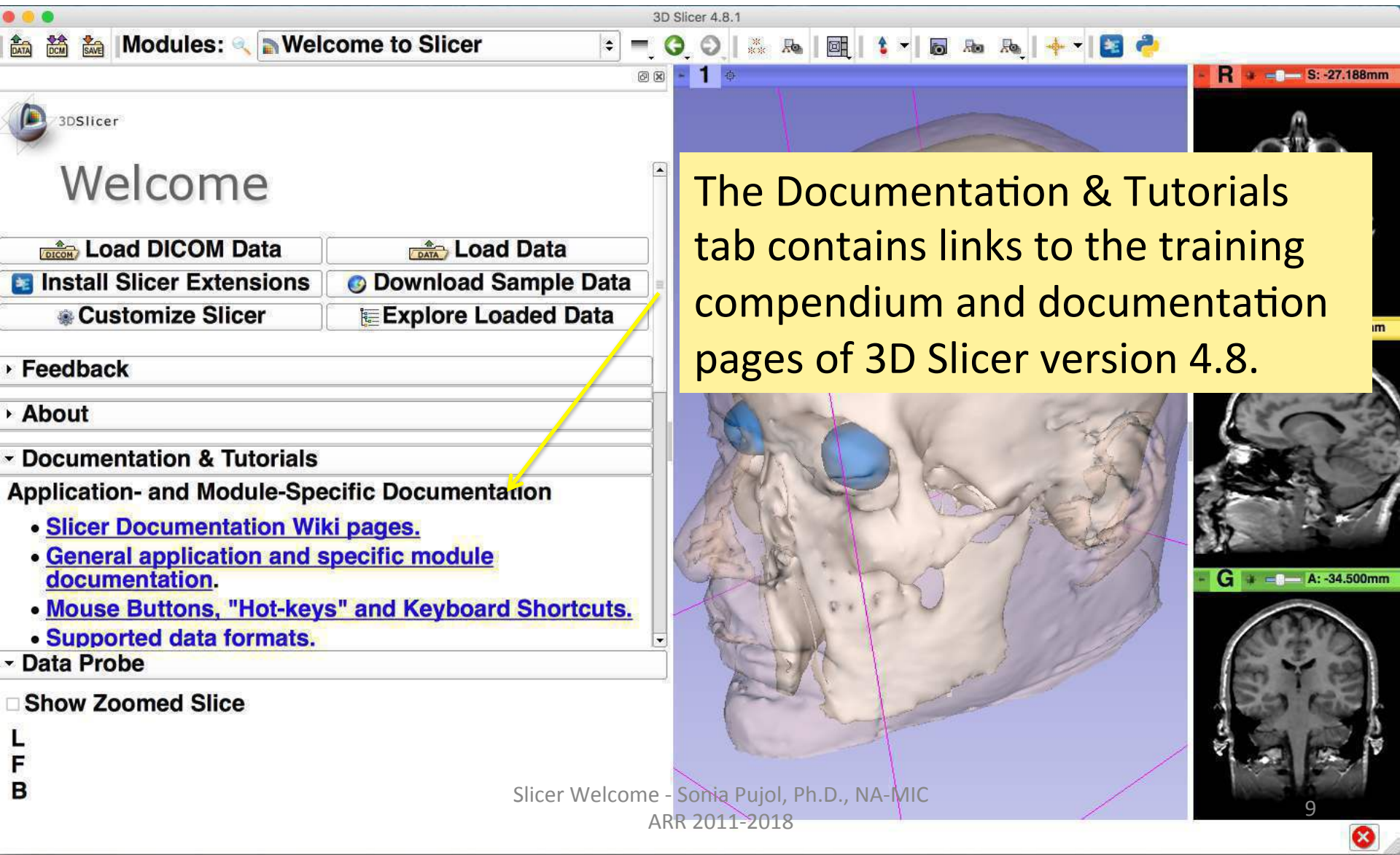
User Interface (UI) panel of the Slicer Welcome Module

3D viewer

Data Probe

2D anatomical viewers

Welcome Module



Slicer 4.8 Training & Documentation

Documentation/4.8/Training

[Home](#) < [Documentation](#) < 4.8 < [Training](#)

Introduction: Slicer 4.8 Tutorials

- This page contains "How to" tutorials with matched sample data sets. They demonstrate how to release) to accomplish certain tasks.
 - For tutorials for other versions of Slicer, please visit the [Slicer training portal](#).
 - For "reference manual" style documentation, please visit the [Slicer 4.8 documentation page](#)
 - For questions related to the Slicer4 Training Compendium, please send an e-mail to [Sonia Pujoi](#)
-
- Some of these tutorials are based on older releases of 3D Slicer. The concepts are still useful but features will be different in updated versions.

Contents [hide]

- 1 Introduction: Slicer 4.8 Tutorials
- 2 Quick Start Guide
 - 2.1 Downloading and Installing Slicer
- 3 General Introduction
 - 3.1 Slicer Welcome Tutorial
 - 3.2 Slicer4Minute Tutorial
- 4 3D Visualization
 - 4.1 Slicer4 Data Loading and 3D Visualization
 - 4.2 Slicer4 3D Visualization of DICOM images for Radiology Applications
- 5 Programming
 - 5.1 Slicer4 Programming Tutorial
 - 5.2 Developing and contributing extensions for 3D Slicer
- 6 Segmentation
 - 6.1 Slicer4 Image Segmentation
- 7 Registration
 - 7.1 Slicer4 Image Registration
 - 7.2 Slicer Registration Case Library
- 8 Slicer Extensions
 - 8.1 Slicer4 Diffusion Tensor Imaging Tutorial
 - 8.2 Slicer4 Neurosurgical Planning Tutorial
 - 8.3 Slicer4 Quantitative Imaging tutorial
 - 8.4 Slicer4 IGT

Documentation/4.8

[Home](#) < [Documentation](#) < 4.8

[Nightly](#) [4.8](#) [4.6](#) [4.5](#) [4.4](#) [4.3](#) [4.2](#) [4.1](#) [4.0](#) [3.6](#) [3.5](#) [3.4](#) [3.2](#) [ALL VERSIONS](#)

Where to start ?

- [Getting started](#)
Quick overview about Slicer
- [Training pages](#)
Information on how to use Slicer 4.8
- [FAQ](#)
Set of common questions/answers
- [Discussion Forum](#) 
The most effective way to get help from the community

How to

- [Report a problem / Create a feature request](#) **NEW**

Slicer Application

- [Installation / Uninstallation](#)
- [Main Application User-Interface](#)
- [Application Settings](#)
- [Extensions Manager](#)
- [Mouse Buttons, "Hot-keys" and Keyboard Shortcuts](#)
- [Recommended Computer Configurations](#)
- [Loading or Saving data and listing of supported data formats.](#)
- [Slicer Lookup Tables](#)

Modules

-  [Annotations](#)
 -  [Data](#)
 -  [Data Store](#)
 -  [DICOM](#)
 -  [Editor](#)
 -  [Markups](#)
 -  [Models](#)
 -  [Scene Views](#)
 -  [Segmentations](#)
 -  [Segment Editor](#) 
 -  [Transforms](#)
 -  [View Controllers](#)
 -  [Volume Rendering](#)
 -  [Volumes](#)
 -  [Welcome to Slicer](#)
- [Modules by category](#)
 - [Modules by name](#)
 - [Modules by contributing organization](#)
 - [Modules by contributing individual](#)
 - [Modules by type](#)
 - [Modules by extension](#)
 - [Extensions by category](#)
 - [Extensions by name](#)
 - [Extensions by contributing organization](#)
 - [Extensions by contributing individual](#)

Developers Corner

- [Information for Software Developers](#) **UPDATED**
Source code, contribute patch, develop modules

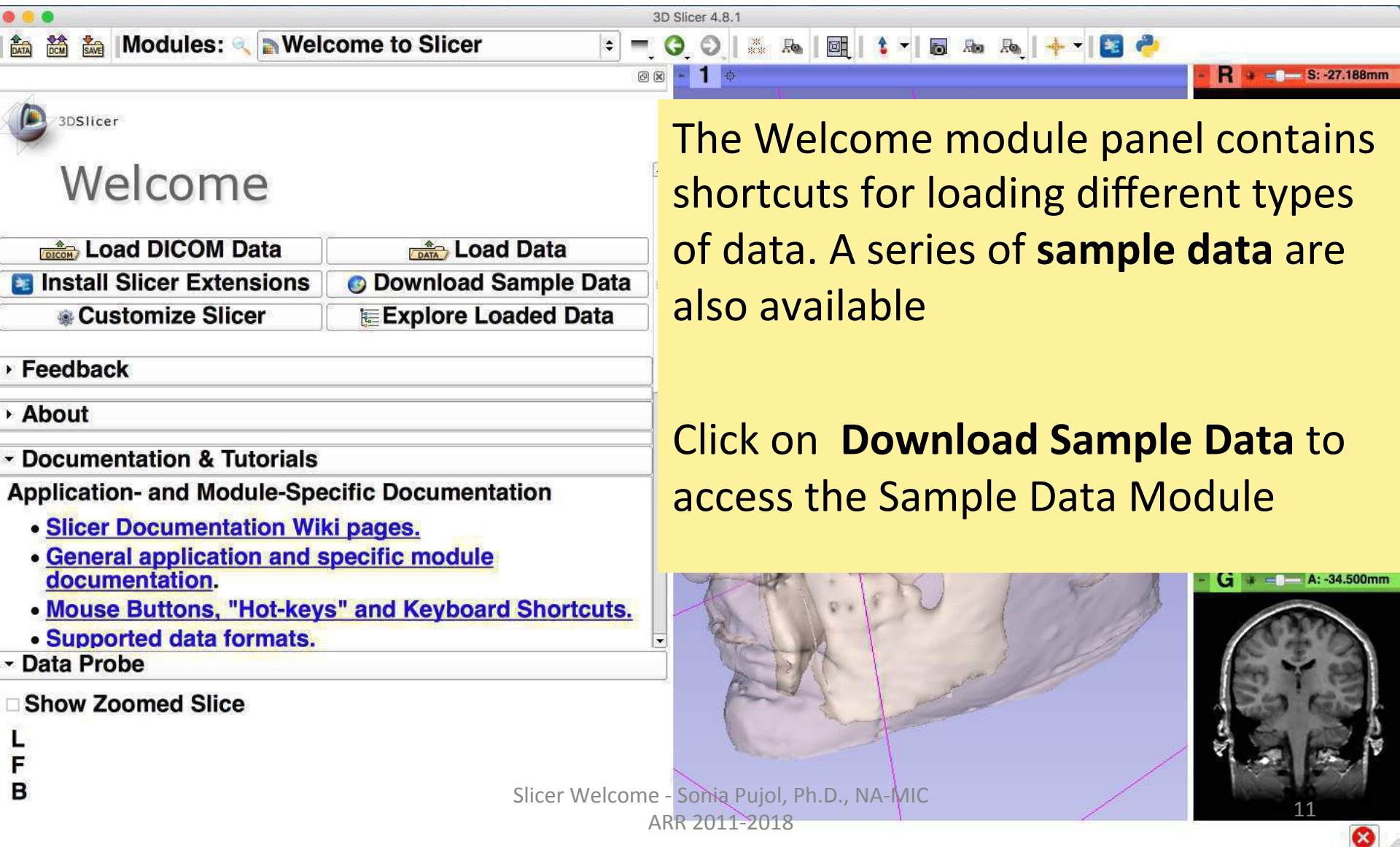
Miscellaneous

- [Documentation guidelines](#)
Slicer user documentation principle and guidelines
- [Visual blog](#)
Set of screenshots showing Slicer in action.
- [Release Notes](#)
Platform specific issues and considerations
- [Announcements & Acknowledgments](#)
- [Registration Library](#) 
Real-life example cases of using the Slicer registration tools, incl. datasets and step-by-step instructions to follow and try yourself.

Documentation in other languages

- [Español](#)

Welcome Module



3D Slicer 4.8.1

Modules: Welcome to Slicer

1 R S: -27.188mm

Welcome

Load DICOM Data **Load Data**

Install Slicer Extensions **Download Sample Data**

Customize Slicer **Explore Loaded Data**

Feedback

About

Documentation & Tutorials

Application- and Module-Specific Documentation

- [Slicer Documentation Wiki pages.](#)
- [General application and specific module documentation.](#)
- [Mouse Buttons, "Hot-keys" and Keyboard Shortcuts.](#)
- [Supported data formats.](#)

Data Probe

☐ Show Zoomed Slice

L
F
B

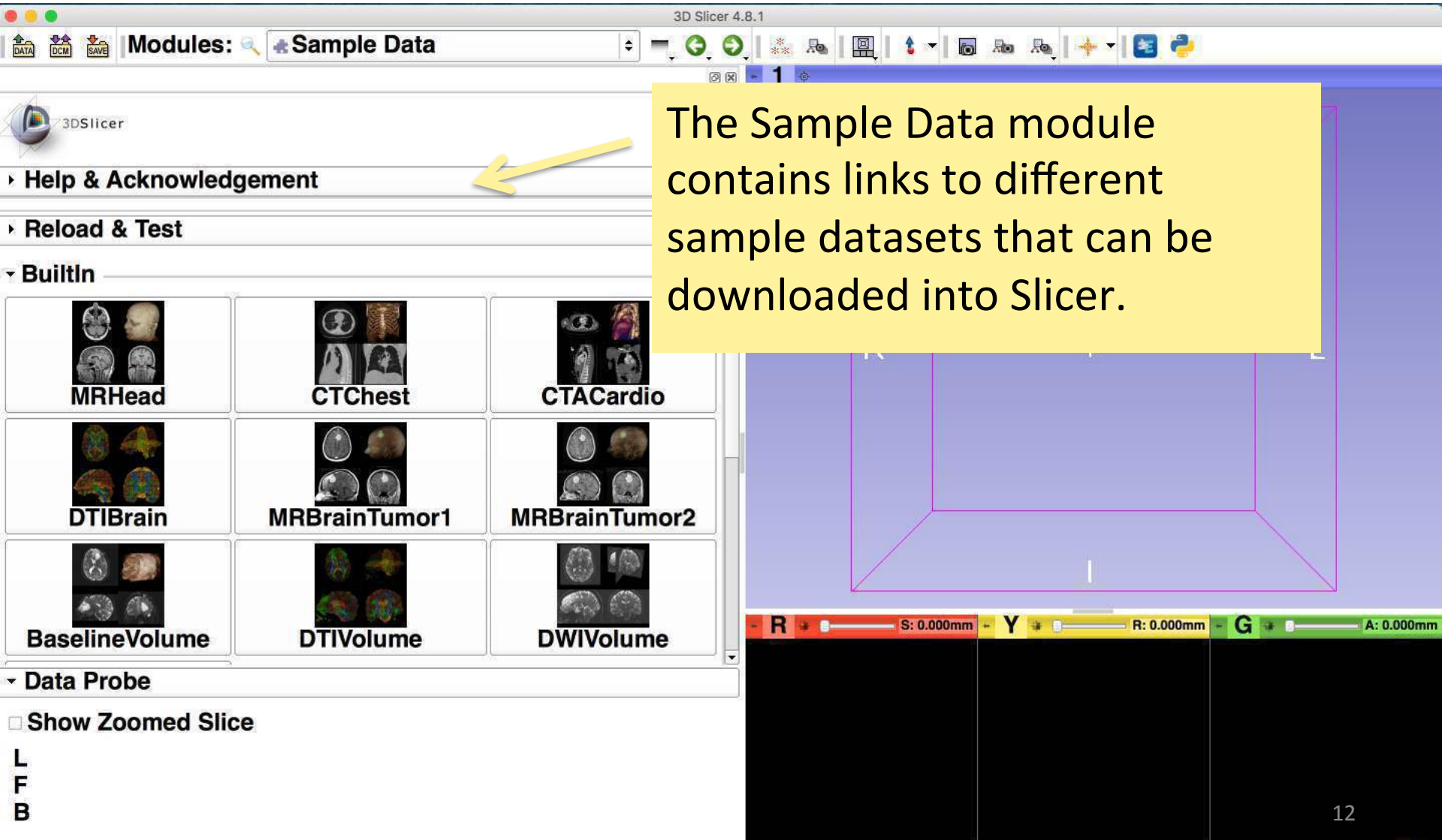
The Welcome module panel contains shortcuts for loading different types of data. A series of **sample data** are also available

Click on **Download Sample Data** to access the Sample Data Module

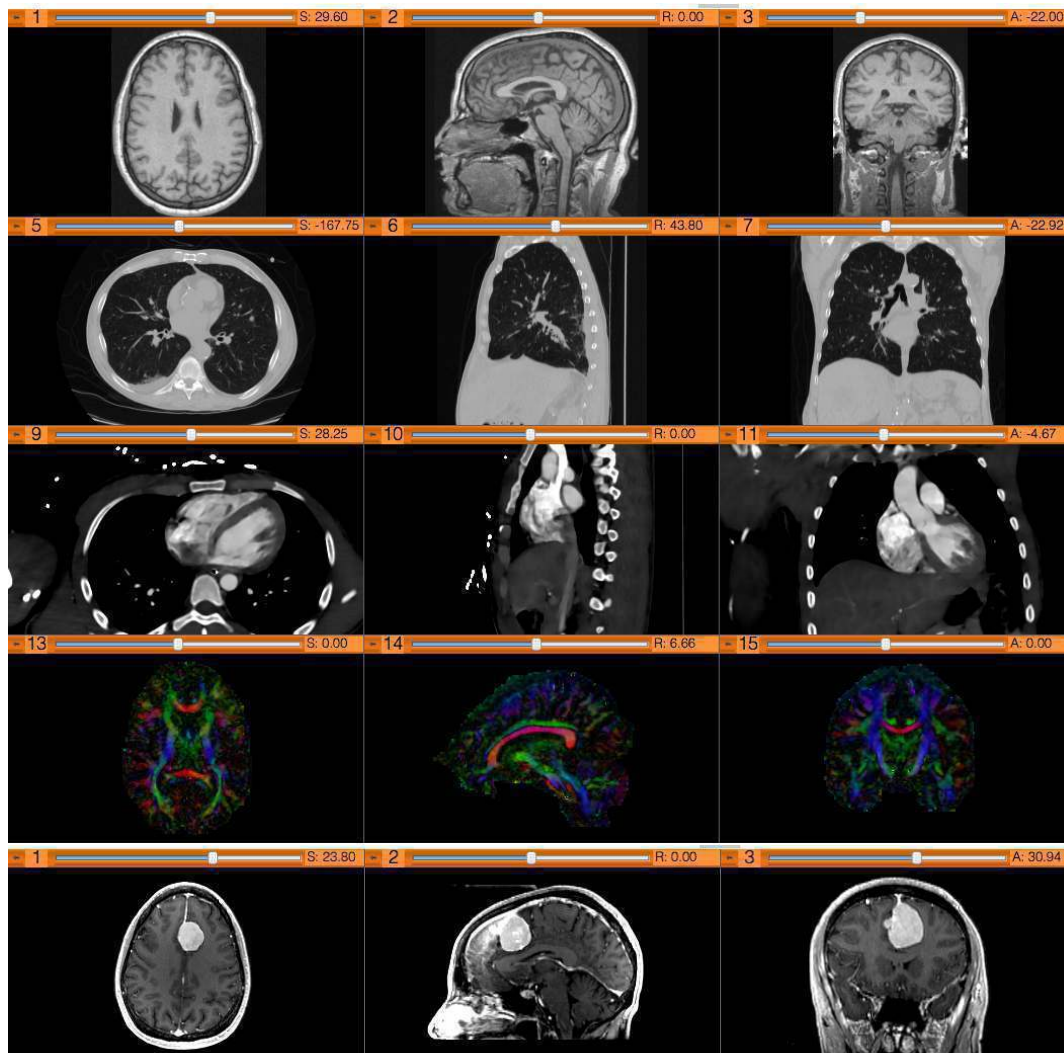
Slicer Welcome - Sonia Pujol, Ph.D., NA-MIC
ARR 2011-2018

11

Sample Data



Sample Data



Brain MRI

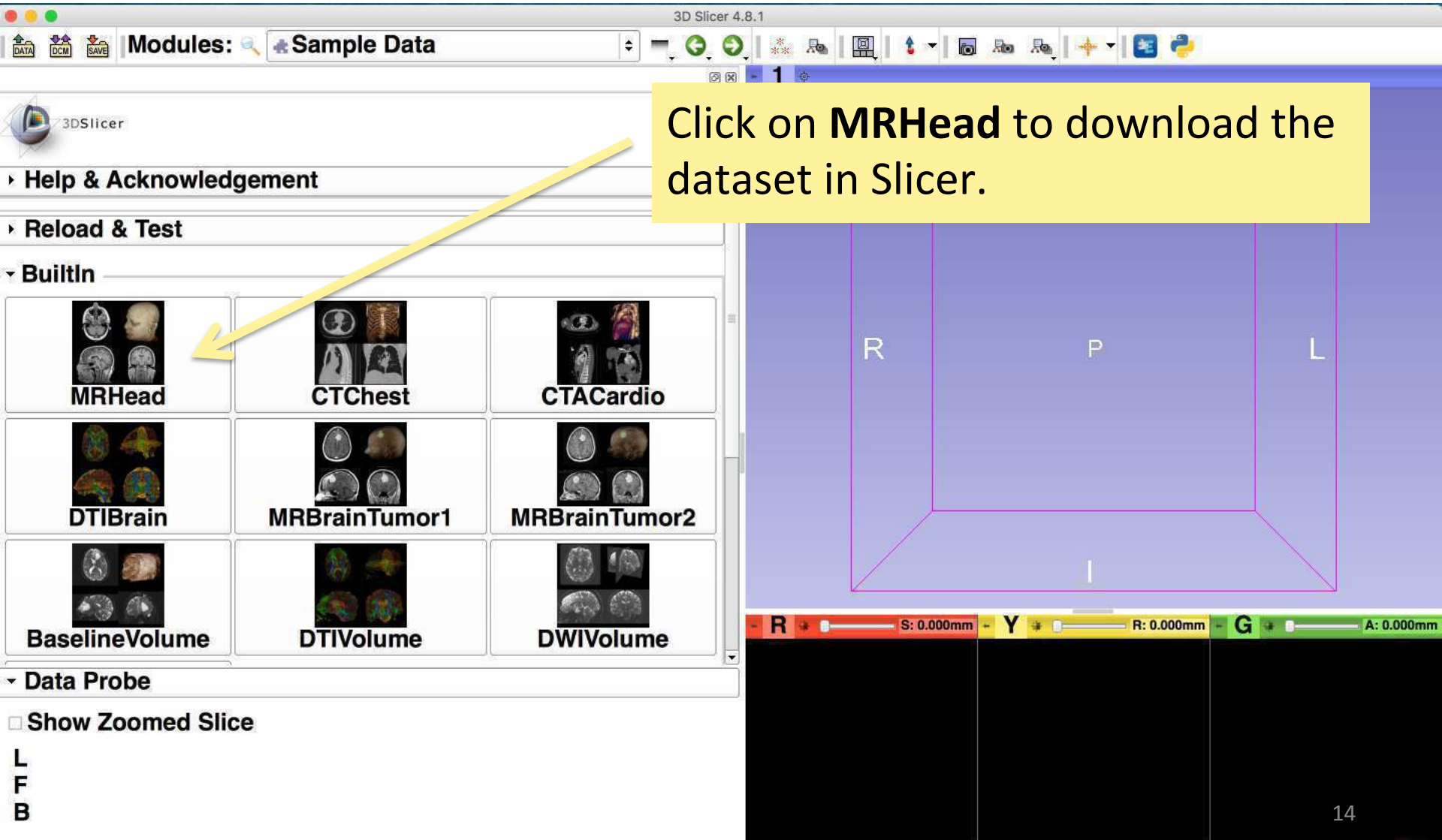
Chest CT

Cardiac CT

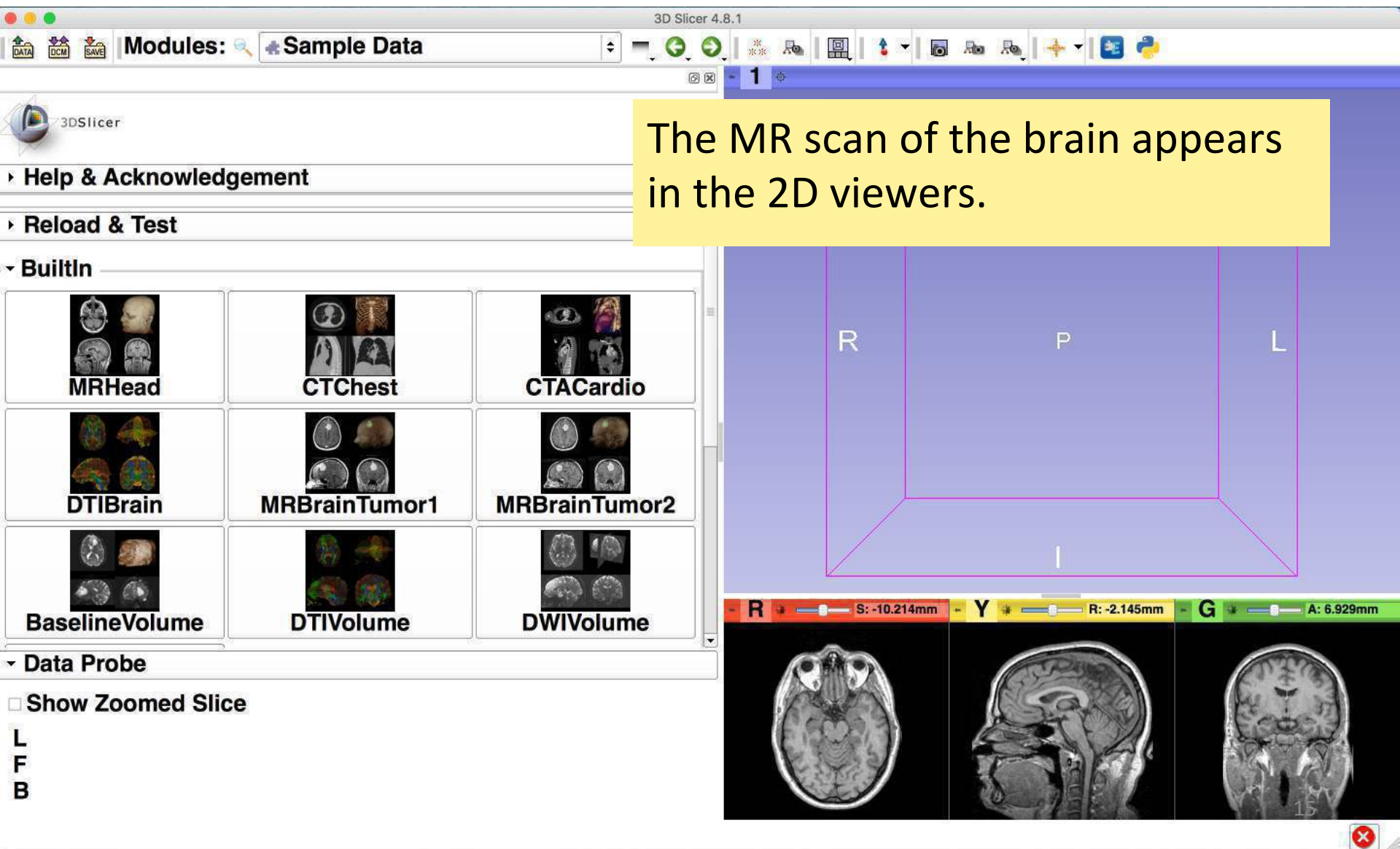
Diffusion Tensor
Imaging (DTI) Dataset

Brain MRI
(tumor patient)

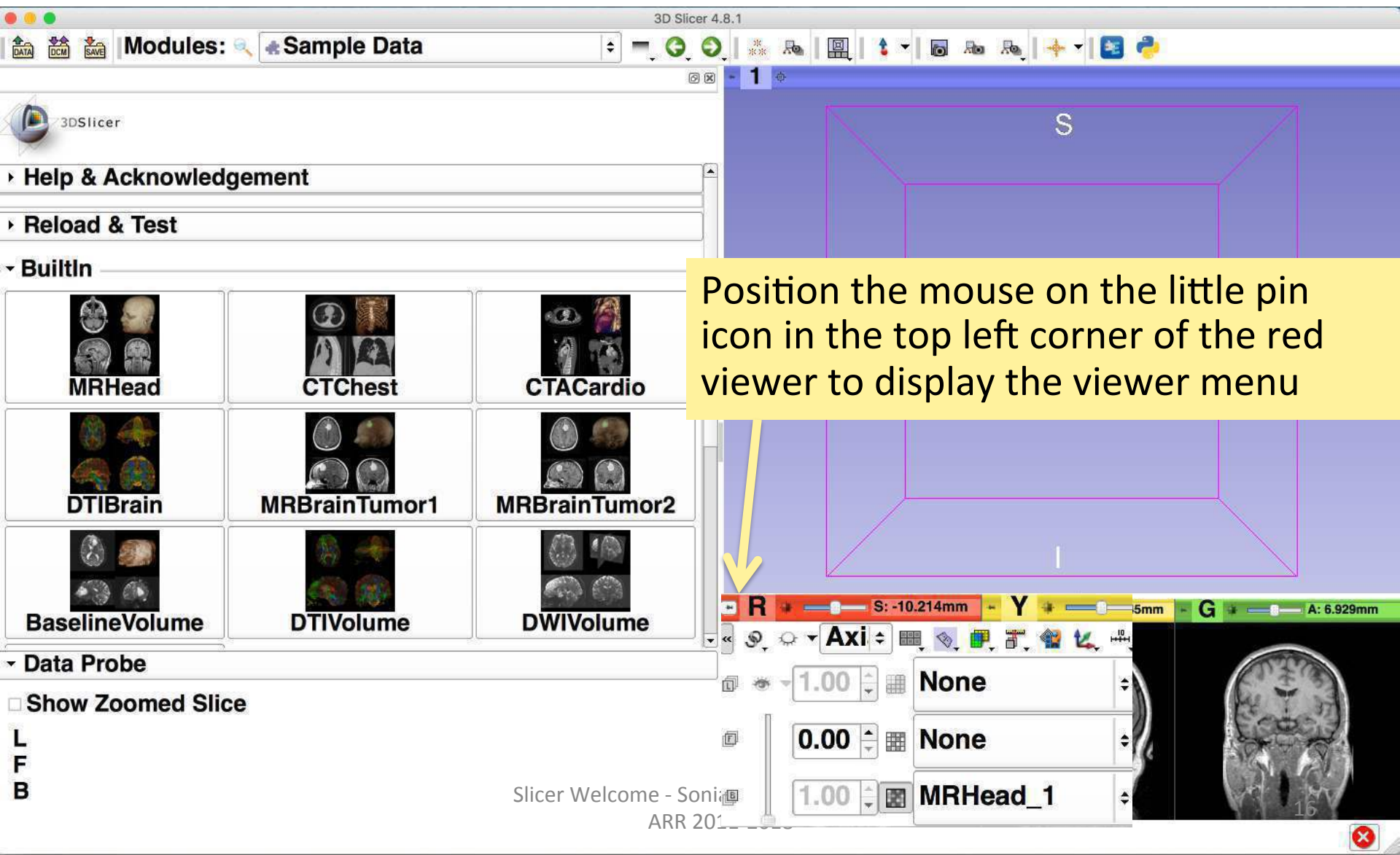
Sample Data



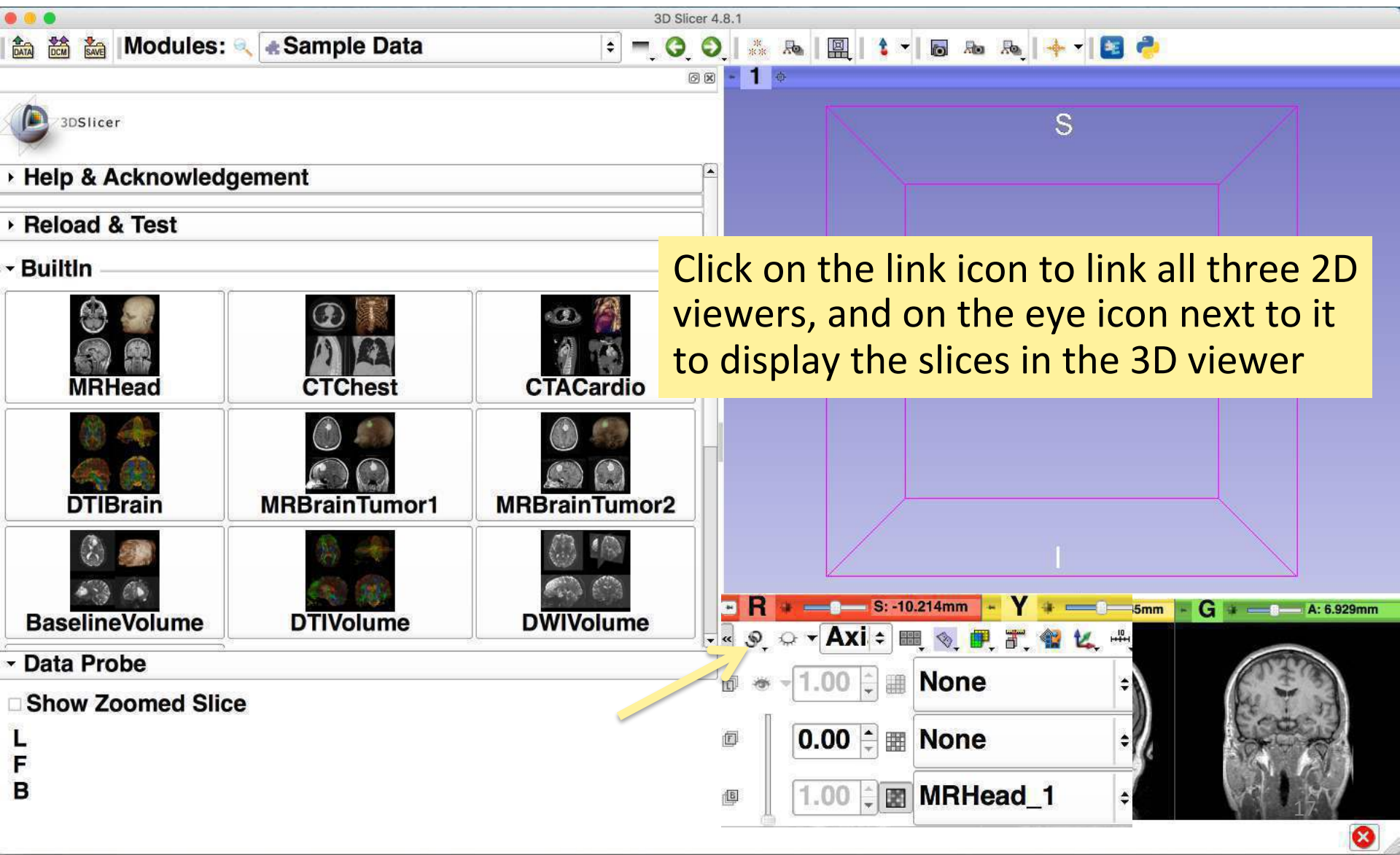
Welcome Module



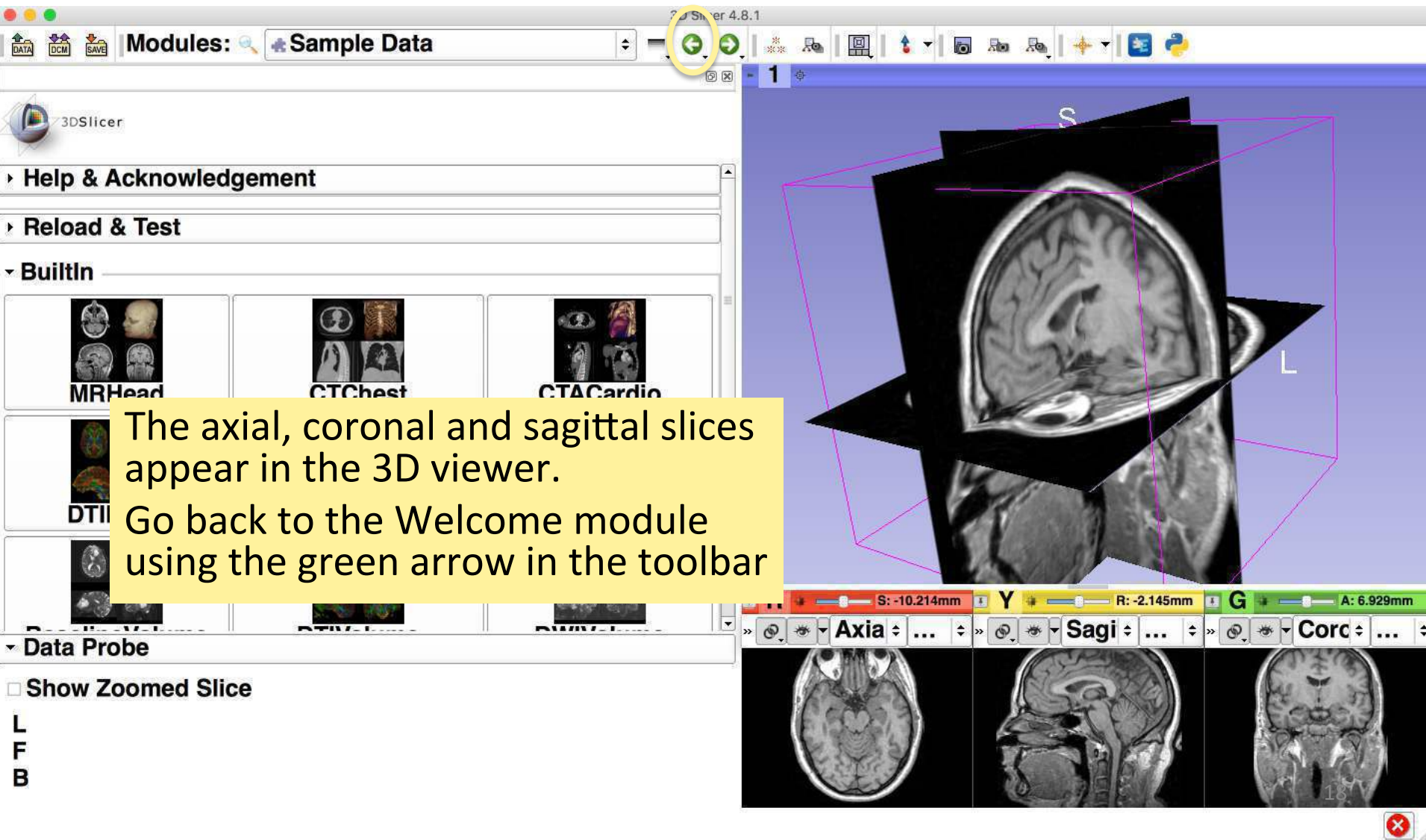
MR Brain Sample Dataset



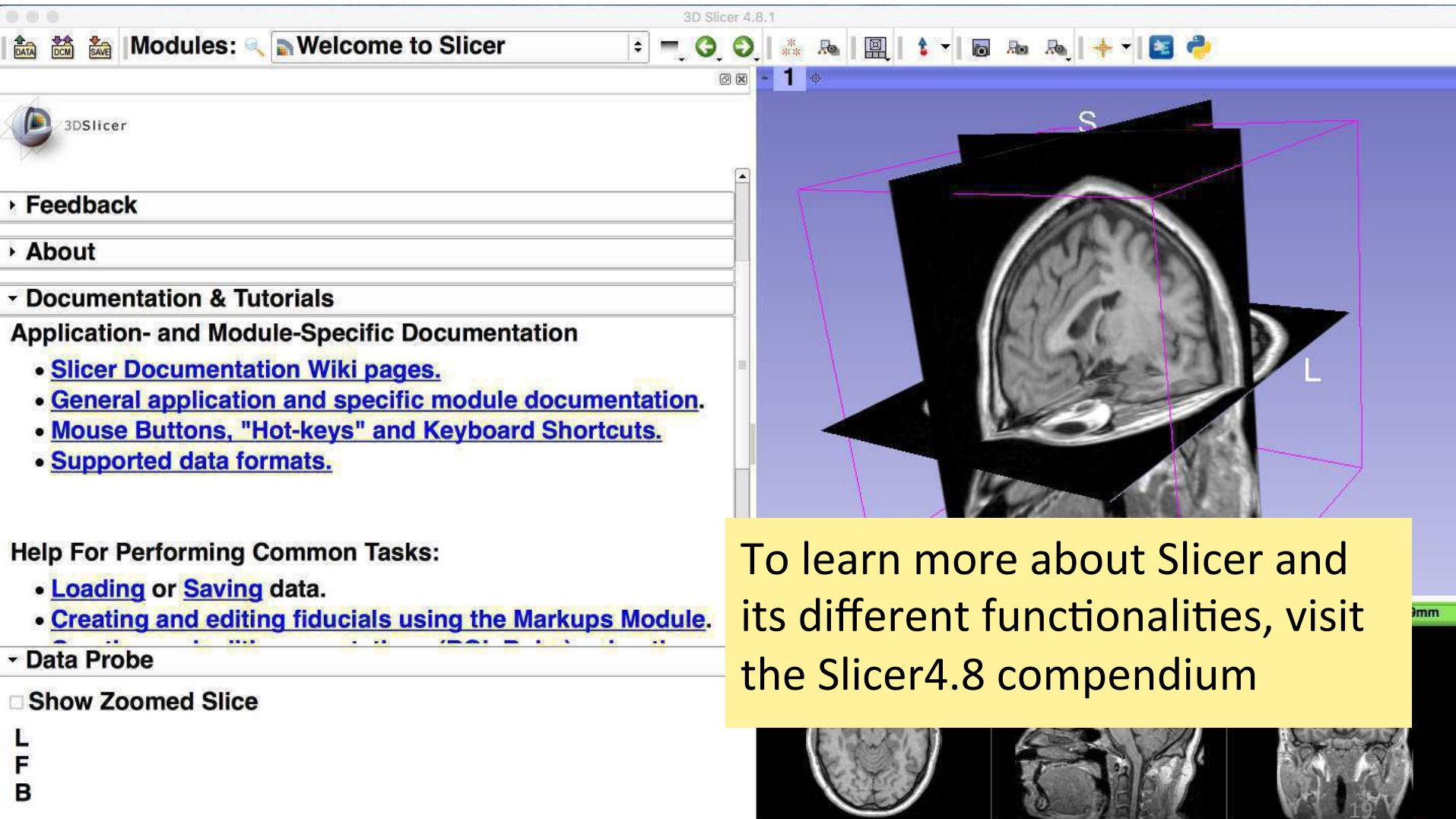
MR Brain Sample Dataset



MR Brain Sample Dataset



Going Further



The screenshot displays the 3D Slicer 4.8.1 software interface. The top toolbar includes icons for Data, DCM, Save, and various viewing and manipulation tools. The left sidebar contains a 'Modules' list with 'Welcome to Slicer' selected, and a 'Documentation & Tutorials' section. The main window shows a 3D view of a brain MRI scan with a yellow callout box containing text. The bottom of the interface shows a 'Data Probe' section with a 'Show Zoomed Slice' checkbox and a 'LFB' label.

3D Slicer 4.8.1

Modules: Welcome to Slicer

Feedback

About

Documentation & Tutorials

Application- and Module-Specific Documentation

- [Slicer Documentation Wiki pages.](#)
- [General application and specific module documentation.](#)
- [Mouse Buttons, "Hot-keys" and Keyboard Shortcuts.](#)
- [Supported data formats.](#)

Help For Performing Common Tasks:

- [Loading or Saving data.](#)
- [Creating and editing fiducials using the Markups Module.](#)

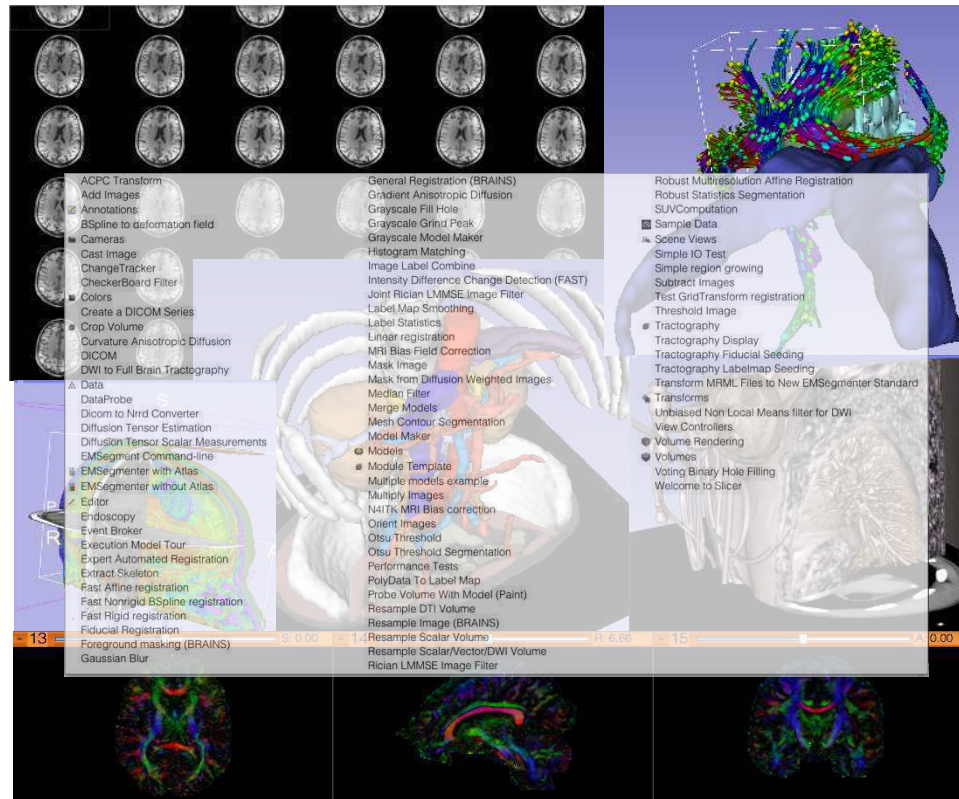
Data Probe

☐ Show Zoomed Slice

LFB

To learn more about Slicer and its different functionalities, visit the Slicer4.8 compendium

Going Further



<http://www.slicer.org/slicerWiki/index.php/Documentation/4.8/Training>

Acknowledgments



National Alliance for Medical Image Computing

NIH U54EB005149



Neuroimage Analysis Center

NIH P41EB015902